# DATA VISUALIZATION FOR DECISION MAKING

## PRACTICAL REPORT

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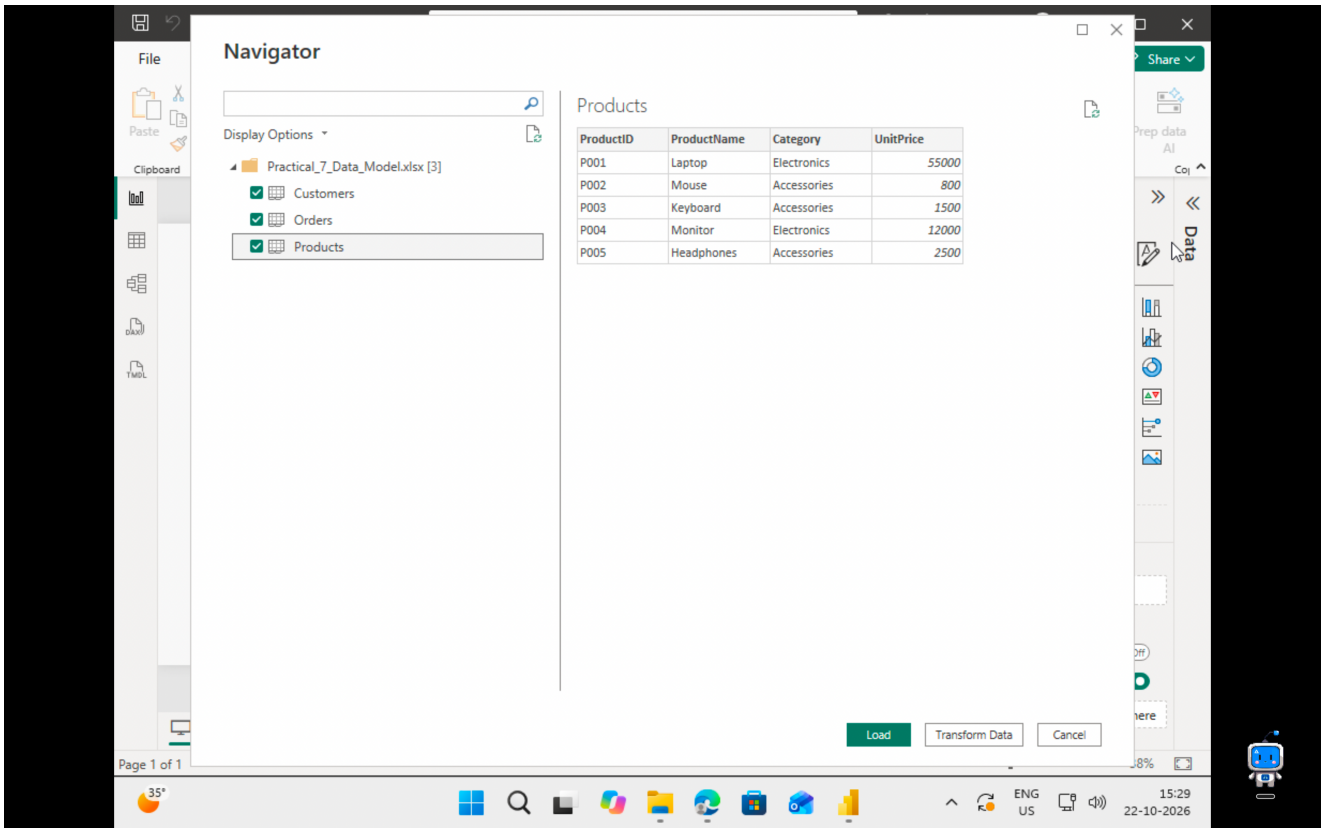
## Practical 7: Build a Data Model

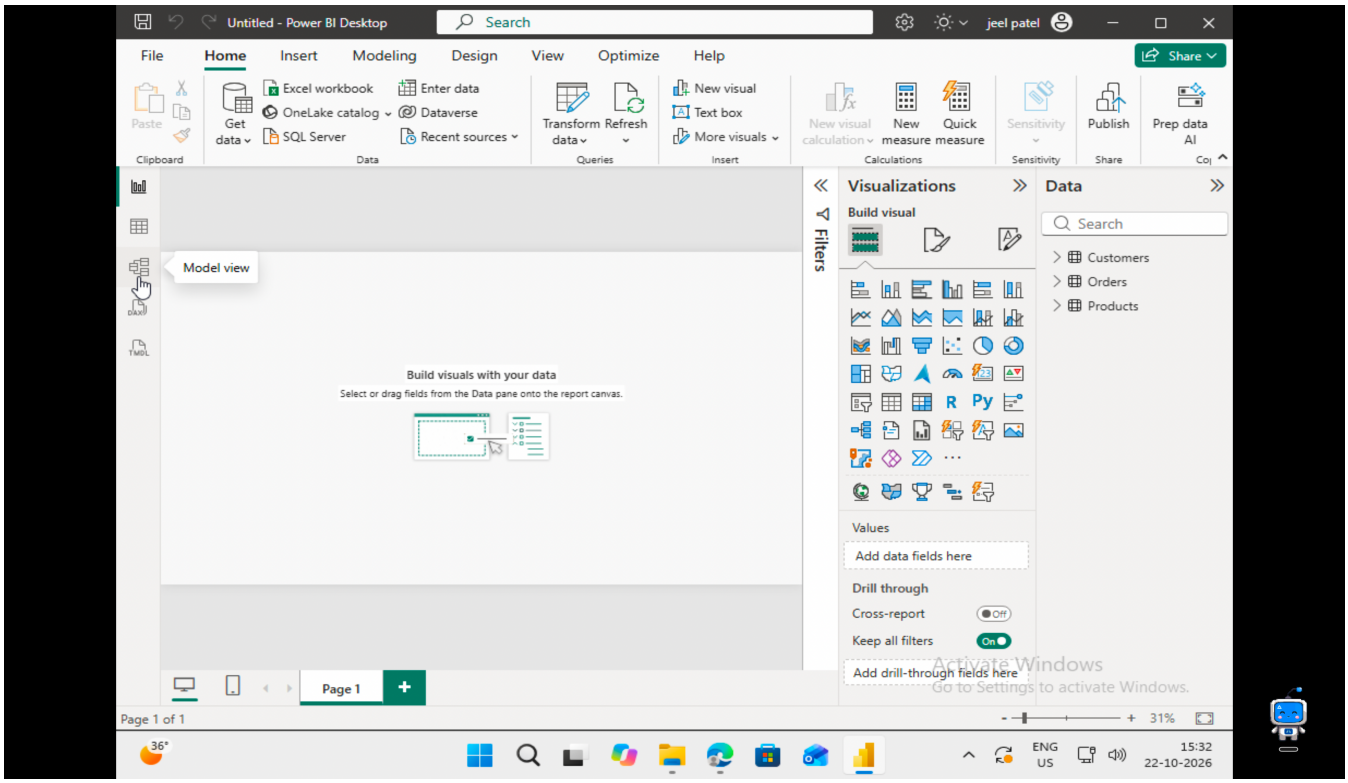
# Practical 7: Build a Data Model

## Step 1: Identify the Tables and Key Columns

• Open the Power BI report containing the tables you want to model, or import the required tables using **Home > Get Data**.

• Identify each table's primary key and any foreign keys. In this example, Customers, Orders, and Products are related using CustomerID and ProductID.



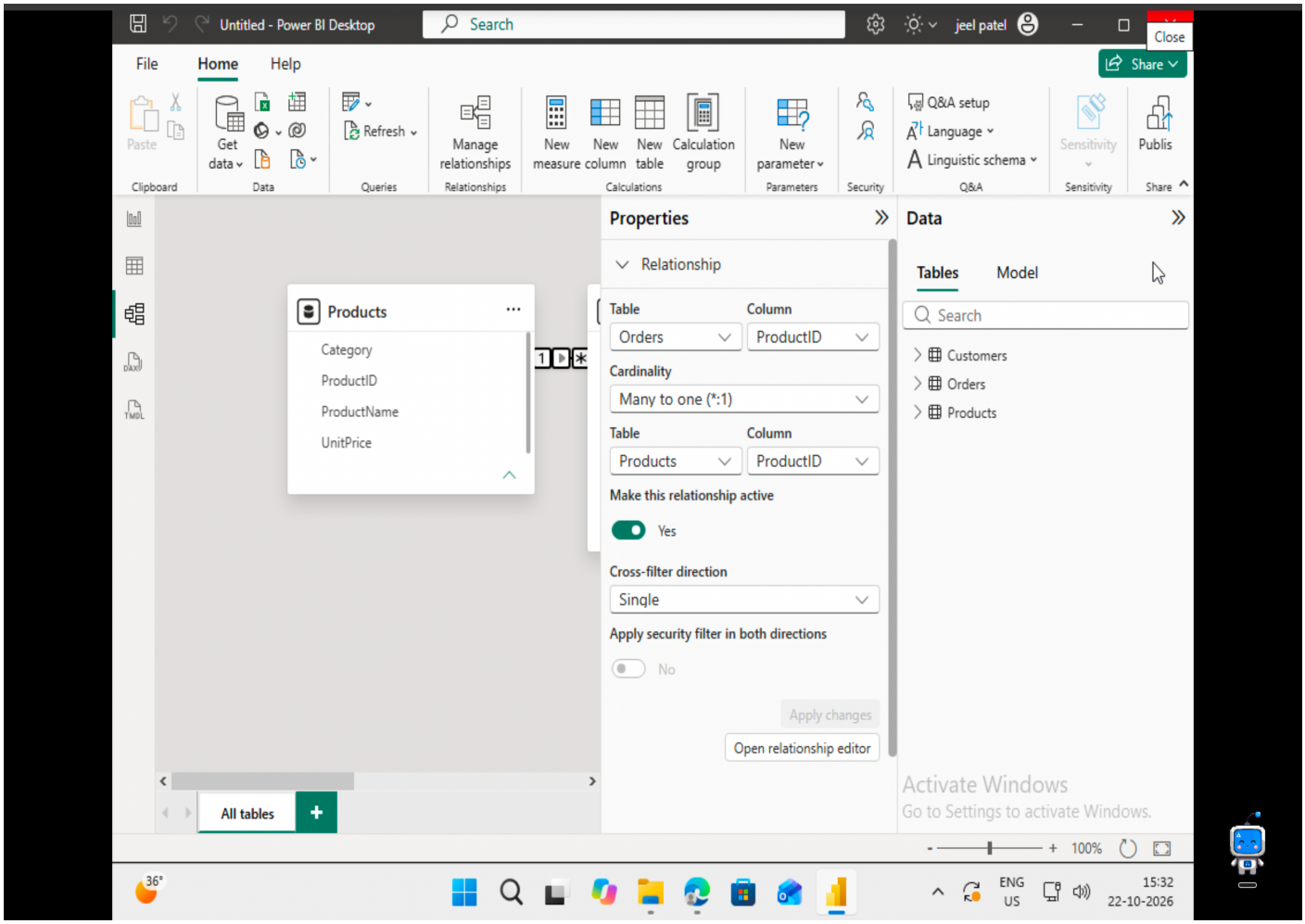



# Practical 7: Build a Data Model

## Step 2: Open Model View

• Select **Model view**from the left navigation pane in Power BI Desktop.

• Review the tables and their columns. Arrange the table cards so the key columns and relationships are easy to inspect.

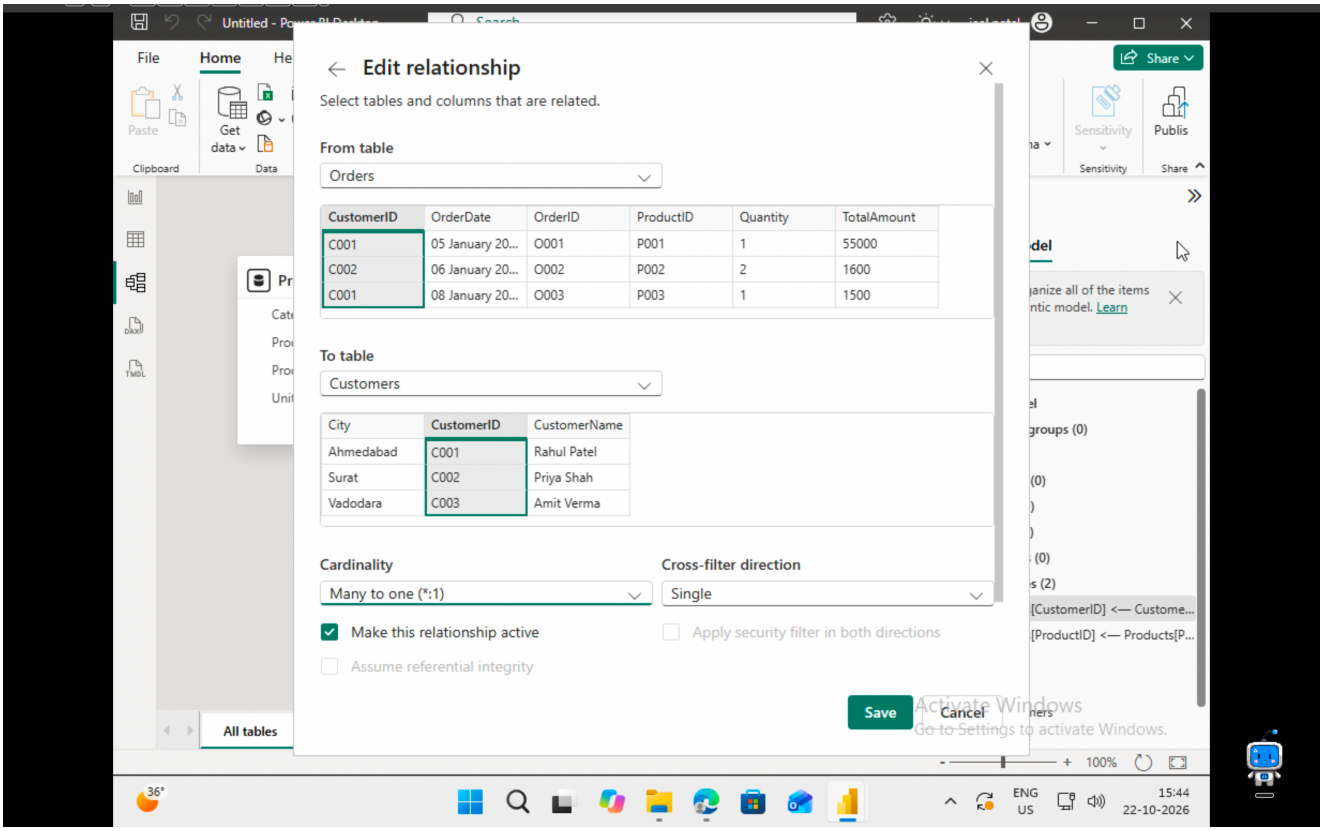


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## Step 3: Create a Relationship

• Drag **Customers[CustomerID]** into **Orders[CustomerID]** or choose **Modeling > Manage relationships > New**

• Select the matching table and column on each side, then set the appropriate cardinality. For unique customer IDs linked to multiple orders, use many to one **( * : 1 )** .

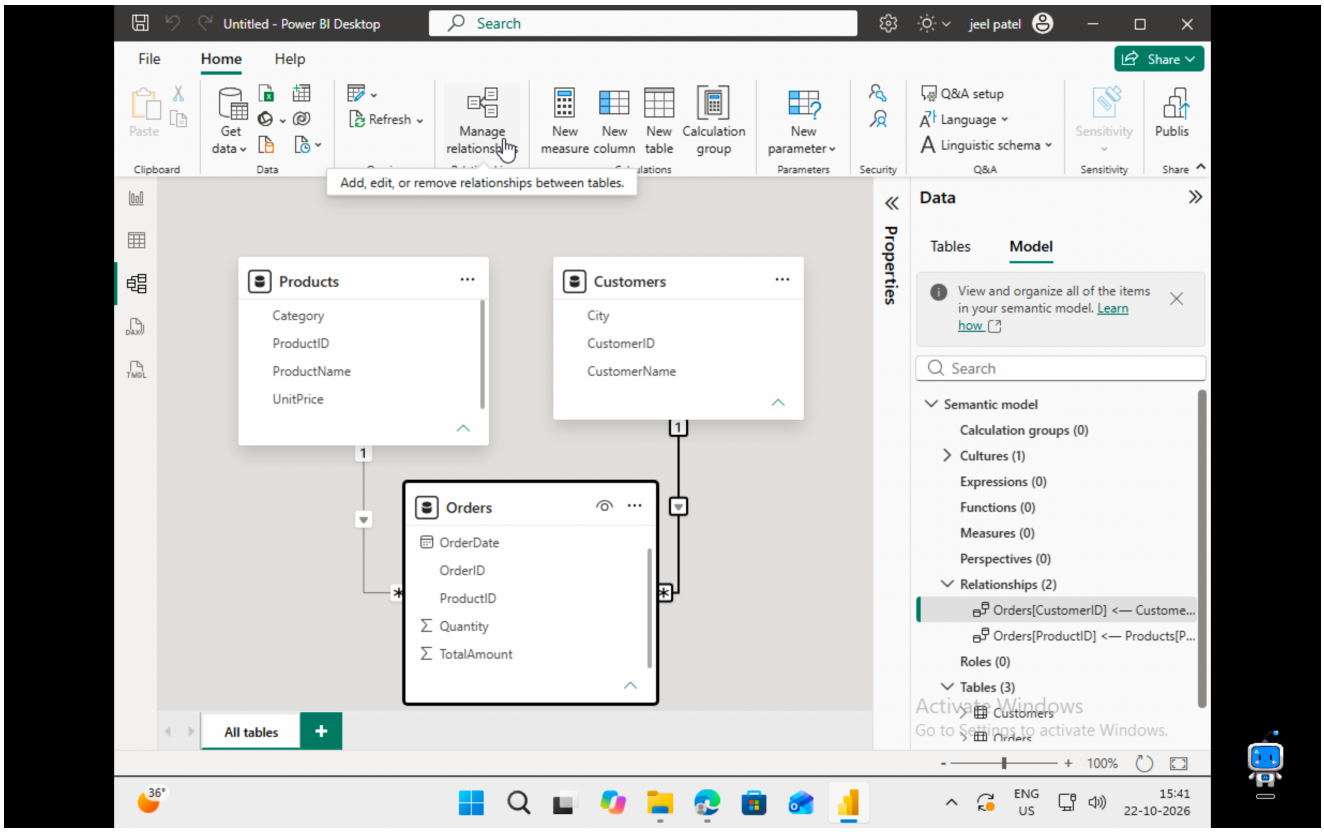


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## Step 4: Verify Relationship Settings

• Confirm that the intended columns are selected and that the relationship is active.

• Use compatible data types in both key columns. The column on the 'one' side should contain unique values; avoid ambiguous or unnecessary relationships.

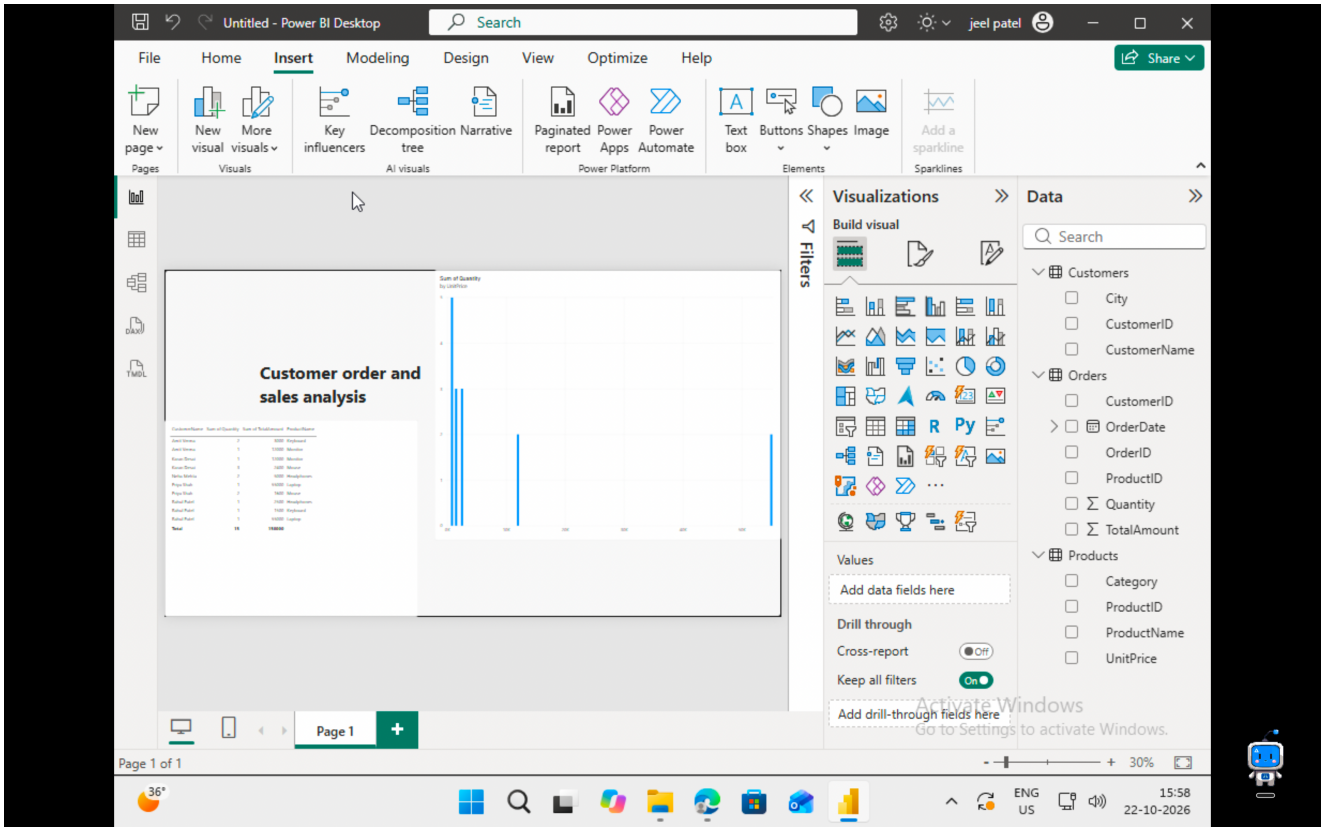


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## Step 5: Test the Model with a Visualization

• Open **Report view**and create a table or chart using fields from related tables.

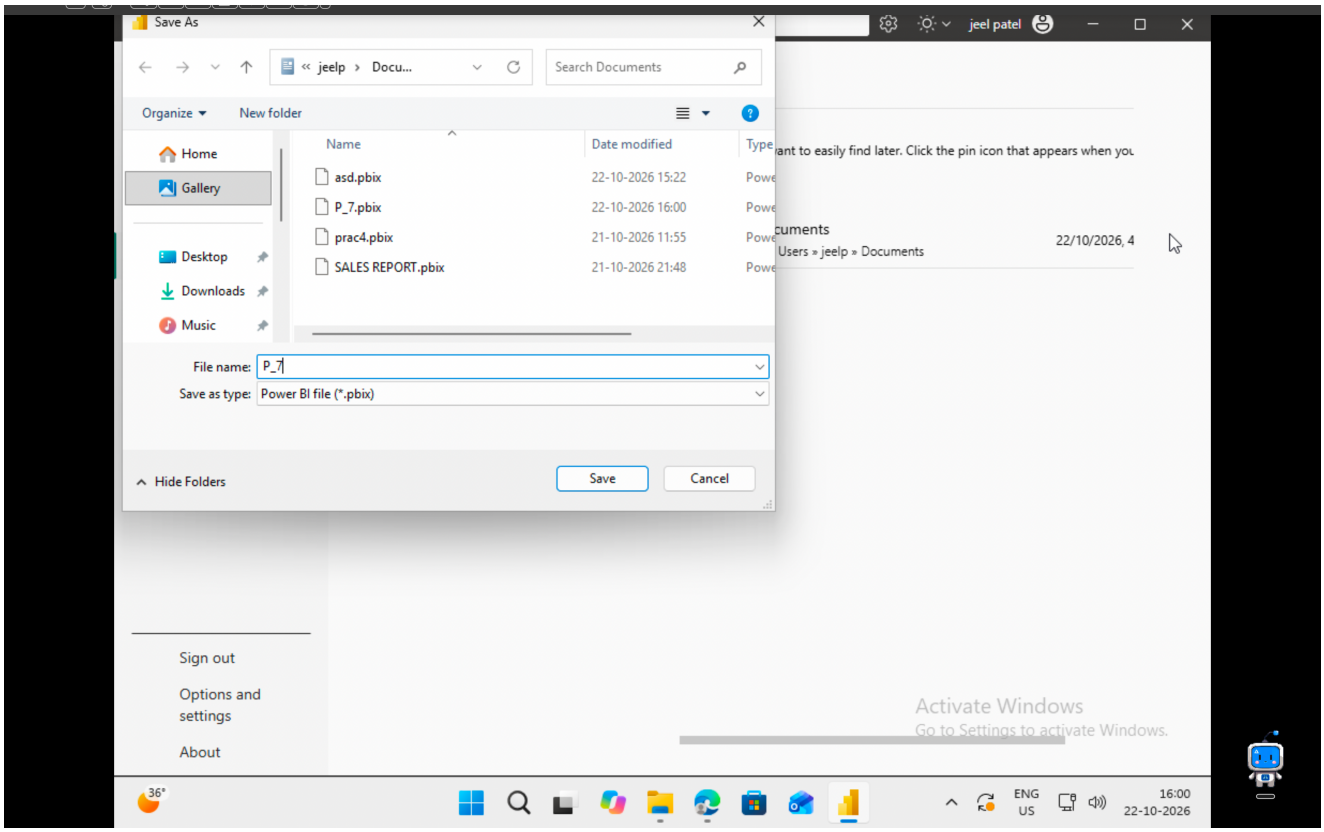
• For example, use CustomerName from Customers and an order count or amount from Orders. Check that the results reflect the relationship correctly.



# Practical 7: Build a Data Model

## Step 6: Review and Save the Completed Model

- Return to **Model view**and check that the relationships form the intended structure.
- Save the report using **File > Save**, for example as **Practical_7_Build_a_Data_Model.pbix**



# Practical 7: Build a Data Model

## Result

Relationships were created between multiple tables in Power BI to form a data model. The relationship settings were reviewed, and a basic visualization was used to test whether fields from related tables worked together as expected.

**Note:** The screenshots in this report are illustrative Power BI interface mock-ups. Table names and example fields are for demonstration; use the actual tables and key columns in your database.